1 Problem Statement

Background. Large vision-language models (LVLMs) have been used to interpret medical images and can produce step-by-step reasoning. Interpreting a medical image roughly involves three stages: looking at the relevant region, correctly recognizing what the region contains, and reasoning from it to a diagnosis. Existing studies show that LVLMs make errors at all three stages. First, models may **not look at the image**: they often answer from language priors, for example giving affirmative answers based on terms that frequently co-occur in the question (such as "lung" and "pneumonia"), and replacing the image region that the question refers to often leaves the answer unchanged. Second, models **misread** images: a physician review of GPT-4V on clinical image challenges found that, even for correctly answered questions, flawed image comprehension was the most common problem in its explanations; on questions about lesions and their locations, mainstream models perform at or below chance. Third, models **reason incorrectly**: the same review found that part of the step-by-step reasoning was flawed even for correctly answered questions, and error analyses of general multimodal reasoning also show that a substantial share of errors stems from reasoning rather than perception. These three types of errors roughly correspond to the division of radiologists' reading errors into search errors, recognition errors, and decision-making errors.

Motivation. An LVLM's reasoning over a medical image typically first makes **visual claims** about the image (e.g., "there is an opacity in the right lower lung") and then draws **clinical inferences** from them (e.g., "the opacity is consistent with pneumonia"). To correct the three types of errors through training, the supervision signal needs to answer three questions: Is the reasoning grounded in the image? Are the visual claims correct? Do the clinical inferences hold?

Visual grounding. Multimodal PRMs take the image as input and assess the consistency of reasoning steps with the image content. This capability forms the basis of multimodal process supervision, on which this work builds.

Correctness of visual recognition. Relevance to the image does not imply correct recognition. A visual claim may be inconsistent with the image in the presence or location of a finding, yet existing PRMs merge this with reasoning quality into a single step score, making recognition-level errors difficult to reflect separately. We therefore construct supervision signals for visual claims based on expert masks, enabling the PRM to judge recognition results independently.

Validity of clinical inference. Even when recognition is correct, the clinical inference drawn from it may not hold. Existing medical PRMs mainly address text-based reasoning, while image-aware verification methods usually evaluate each sentence as a whole; neither separates the validity of an inference from the visual premises it relies on. We evaluate inferences separately with retrieved medical knowledge, and treat an inference as dependent when its premise is incorrect, so that the same error is not counted twice.

Research questions.

- **Question 1:** For each step of medical image reasoning, can we judge separately whether it is grounded in the image, whether its visual recognition is correct, and whether its clinical inference holds?

- **Question 2:** When these typed signals are used as step-level rewards in reinforcement learning, can they make the model rely more on the image, reduce misreading and faulty reasoning, and improve answer accuracy? We compare against GRPO with answer rewards only and against PRM rewards that use a single step score.

3 Method

3.1 Overview

We build a typed process reward model for medical image reasoning. The method consists of two kinds of evaluation signals and two uses.

Evaluation signal	Question it answers	Main source of supervision
Visual fact signal	Is the visual claim consistent with verifiable image facts?	Expert masks and their annotation scope, completeness, and spatial information
Conditional inference signal	Assuming the listed premises hold, is the current conclusion sufficiently supported?	Retrieved medical knowledge and human-verified annotation rules

The two signals are linked through the dependency between an inference and its premises: the visual evaluation checks the observations in the case, and the inference evaluation checks the relation from observations to conclusion. If a visual claim on which an inference relies has been judged incorrect, the inference is labeled "dependent" and is no longer counted separately as an inference error.

The trained PRM has two uses:

1. **Best-of-N selection:** without updating the policy model, score multiple candidate reasoning chains and select the highest-scoring response.
2. **Reinforcement learning:** convert the PRM's judgments into step-level rewards to train the policy model.

We treat the accuracy of the PRM's judgments and the Best-of-N performance as the primary evaluation, and on this basis evaluate the effect of reinforcement learning.

3.2 Relation to Existing PRMs

(To be completed: survey and summarize the differences from existing PRMs.) For example, Med-PRM verifies steps based on retrieved knowledge but handles text only. Our method can judge whether a failed step is a reasoning error or a visual error.

3.3 Constructing PRM Training Data

Goal: For each reasoning step written by the policy model, automatically generate two kinds of labels: whether the visual claim is correct (correct reading / misreading) and whether the clinical inference holds (valid / invalid reasoning). These labels are used to train the PRM.

We use one example throughout: an abdominal CT image with the question "Is the liver abnormal?". The training set provides masks for this image, marking the liver and a mass within the liver.

3.3.1 Step 1: Generating Reasoning

Procedure: Use the policy model (e.g., Qwen2.5-VL-7B) to sample 8 reasoning chains per question at temperature 0.7. Multiple samples are drawn to obtain different ways of reading the image and different reasoning paths, including both correct ones and ones containing errors, so that the PRM has both positive and negative samples to learn from.

Example prompt:

```
Answer the question based on the image. Reason step by step, with each step
on its own line starting with "Step k:".
When describing the image, state the region you examine and what you see
and do not see there.
Start the last line with "Answer:".
Question: {question}
```

Why ask for what is not seen:

1. Radiology reports routinely state negative findings (e.g., "no pneumothorax"), and such information is useful for diagnosis;

2. Without this prompt, the model rarely states negative findings on its own, so negative claims are too scarce in the training data for the PRM to learn to judge them;

3. It distinguishes "misreading" from "omission": if the model writes "no mass is seen" while a mass is present, it considered the mass but misread the image; only if it never mentions the mass is it an omission.

What is saved: the question ID, image ID, full reasoning text, final answer, and whether the answer is correct for each reasoning chain.

Example:

Step 1: The liver is visible in the image.
Step 2: There is a low-density mass in the right lobe of the liver.
Step 3: No ascites is seen.
Step 4: Therefore, the liver is abnormal; a liver tumor is suspected.
Answer: Yes

3.3.2 Step 2: Parsing Steps

Procedure: Use a large model (e.g., a large Qwen-VL model) to split each step into "visual claims" and "inferences." It **only parses and does not judge correctness**. Correctness is determined in Steps 3 and 4, so that the labels are not affected by the parsing model's own judgment, and so that we can tell whether a failed step is a visual error or a reasoning error.

Each visual claim records four items: finding category, present or absent, location, and whether it is hedged. Each inference records which visual claims it relies on and what its conclusion is. The model is required to **copy the original text spans verbatim**, so that labels can later be mapped back to the original sentences.

Example parsing result (saved as JSON in practice):

Step	Type	Content
1	Visual claim	liver, present
2	Visual claim	mass, present, location: right lobe of the liver
3	Visual claim	ascites, absent
4	Inference	based on Step 2 → the liver is abnormal

3.3.3 Step 3: Labeling Visual Claims (LVLM + Bounding Boxes + Labels)

Procedure: Use a larger LVLM (e.g., Qwen2.5-VL-72B) as the annotator, and have it judge each visual claim against the expert annotations.

Input to the LVLM:

1. **the image**;
2. **the question**;
3. **the reasoning to be judged** (the original text of the visual claim);
4. **expert annotations**: the bounding box and category label of each annotation (the bounding box is the bounding rectangle of the mask).

Judgment rules:

Case	Label
Says "present"; present in the annotations at a consistent location	Supported
Says "present"; absent from the annotations, or at a wrong location	Refuted
Says "absent"; indeed absent from the annotations	Supported
Says "absent"; present in the annotations	Refuted
Hedged statement (e.g., "cannot be excluded"), or cannot be judged from the annotations	Unverifiable

Example prompt:

```
You are a radiology annotation checker. The following are expert annotations;
treat them as ground truth.
Expert annotations: liver [x1,y1,x2,y2]; mass [0.52,0.30,0.68,0.45]
Question: Is the liver abnormal?
Claim to judge: "There is a low-density mass in the right lobe of the liver"

Judgment rules:
- Claimed present: the finding is in the annotations, and its location and
  description match the image -> Supported; otherwise -> Refuted
- Claimed absent: the finding is not in the annotations -> Supported;
  it is in the annotations -> Refuted
- Hedged statement (e.g., "cannot be excluded"), or cannot be judged from
  the annotations -> Unverifiable

Output: Supported / Refuted / Unverifiable, with a one-sentence reason.
```

3.3.4 Step 4: Labeling Inferences (Checking Whether the Inference Is Correct)

Procedure:

- **All claims the inference relies on are "Supported":** retrieve medical knowledge (reusing the retrieval corpus and retriever of Med-PRM), give the premises, the conclusion, the ground-truth answer, and the retrieved passages to a large model, and ask: "Assuming these premises hold, does the conclusion hold?" The output is **Valid** or **Invalid**. Each inference is queried 3 times and the majority is taken.
- **Some claim it relies on is "Refuted":** label it directly as **Dependent**, not as an inference error. The error has already been recorded on that visual claim and is not counted again.

Key points of the judgment prompt: state explicitly that "the premises have been verified; judge only the reasoning relation"; provide the retrieved knowledge; allow only "Valid" or "Invalid" as output.

Example: Step 2, on which Step 4 relies, is "Supported"; "a mass in the liver → the liver is abnormal" holds, so Step 4 is labeled "Valid." If Step 2 were "Refuted," Step 4 would be labeled "Dependent."

3.3.5 Resulting Training Sample

```
Input:  image + question + retrieved knowledge + visual tool output + Steps 1-4
Labels: visual    [Supported, Supported, Supported, None ]
        inference [None,      None,      None,      Valid]
        omission  Complete
```

"None" means the step has no corresponding content (e.g., Step 4 has no visual claim).

Two notes:

- **Masks are used only for labeling and are not part of the PRM's input.** Test images have no masks, and the inputs at training and test time must be the same.
- **What the visual tool output is:** a segmentation model (e.g., nnU-Net) trained on the training-set masks. It takes an image as input and outputs whether each category is present and its approximate location, which is converted to text as a reference for the PRM, e.g., "liver present (0.98); mass possibly present (0.62, right); ascites absent (0.04)." It needs only the image, so it can also be used at test time. The segmentation model must be **trained in 5 folds split by image**: the output for each training image comes from the fold model that has not seen that image; otherwise it would be nearly perfect on training images, and the PRM would learn to trust it blindly.

3.4 Training the PRM

Input format (following Med-PRM in placing label positions after each step):

```
[image]
Question: Is the liver abnormal?
Retrieved knowledge: [1] ... [2] ...
Visual tool output: liver present (0.98); mass possibly present (0.62, right);
                    ascites absent (0.04)
```

```
Step 1: The liver is visible in the image.  Visual:<tag> Inference:<tag>
Step 2: There is a low-density mass in the right lobe of the liver.
                                            Visual:<tag> Inference:<tag>
Step 3: No ascites is seen.                 Visual:<tag> Inference:<tag>
Step 4: Therefore, the liver is abnormal; a liver tumor is suspected.
                                            Visual:<tag> Inference:<tag>
Omission:<tag>
```

Output: at each label position, predict one label from the candidates for that position.

Label position	Candidate labels
Visual	Supported / Refuted / Unverifiable / None
Inference	Valid / Invalid / Dependent / None
Omission	Omission / Complete

Training settings:

- Backbone: Qwen2.5-VL-7B + LoRA ($r = 16$);
- Loss: the sum of cross-entropy losses over all label positions; error classes usually have fewer samples and are given larger weights;
- During training, the visual tool output or the retrieved knowledge is randomly removed from about 20% of the samples, so that the PRM does not rely on only one of them;
- Learning rate 1×10^{-4}, effective batch size 16, 1–2 epochs;
- About 20% of the training set is held out by image for validation, and the best checkpoint is selected by judgment accuracy on the validation set.

Simple check after training: test again with the validation images replaced by blank images. If the accuracy of visual judgments barely changes, the PRM is mainly guessing from the text, and samples that pair the same sentence with different images and opposite labels should be added before retraining.

3.5 How the Trained PRM Supervises the Policy Model

At test time there are no masks; the PRM judges using only the image, the retrieved knowledge, and the visual tool output.

3.5.1 Use 1: Best-of-N Answer Selection (Main Use)

The policy model writes 8 reasoning chains → the PRM scores each one → the highest-scoring one is selected.

3.5.2 Use 2: As a Reward for Reinforcement Learning, to Improve the Diagnosis of Medical LVLMs

Goal: Use the PRM's step-level judgments to train the policy model to read images more accurately and reason more correctly, while maintaining answer accuracy. GRPO is modified as follows: **the answer reward determines how good a whole response is, and the PRM's judgments determine which steps in the response are rewarded or penalized.**